Basal mixing model: forward dynamics and sampling strategies

BasalMixing repository

June 3, 2026

Abstract

This document describes the 1-D basal mixing model used in the BasalMixing repository to interpret deep-ice ^{81}Kr and δ^{40}Ar$_{\mathrm{atm}}$ measurements at GISP2, and the three Markov-chain Monte Carlo (MCMC) strategies the code provides for inferring the mixing parameters. The three strategies (`:profile`, `:sampled`, `:marginal`) differ in how they handle a single nuisance quantity: the time the mixing system has been running before the measurement. We argue that the marginal-likelihood strategy is the correct Bayesian treatment of this problem and is what should be used for publication-quality inference; the other two strategies are kept for backward compatibility and diagnostic comparison.

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1 The basal mixing model

1.1 Physical setting

GISP2 retrieves ice from the bottom of the Greenland Ice Sheet. The deepest ∼13 m of the core sits above the bedrock and, unlike the rest of the column, is thought to have been mechanically and diffusively mixed with material below. Specifically: above $z = z_{\mathrm{lim}} \approx 3040\,\mathrm{m}$ the ice is treated as undisturbed ("clean" ice); below z_{lim}, ice is increasingly contaminated by debris-rich ("dirty") material that acts as a sink/source for the tracers we measure. The model treats this lower layer as a 1-D advection–diffusion column extending to a fixed bedrock at $z_{\mathrm{bed}} = 3053.44\,\mathrm{m}$.

1.2 State variables and grid

The model carries two state variables at every cell centre z_j in a finite-volume discretization:

- $c_{81}(z,t)$: relative ^{81}Kr concentration, dimensionless. Initialized to 1 at $t = 0$ (modern abundance) at every depth.
- $c_{40}(z,t)$: $\delta^{40}\mathrm{Ar}_{\mathrm{atm}}$ concentration, in $\mathrm{cc\,m^{-3}}$. Initialized to the modern equilibrium value $A_{40}^0 = (100\,\mathrm{cm})^3 \cdot a_{\mathrm{tac}} \cdot f_{\mathrm{atm}} \approx 8000\,\mathrm{cc/m^3}$, where $a_{\mathrm{tac}} = 0.08$ is the typical air content and $f_{\mathrm{atm}} = 0.00934$ is the present-day atmospheric $\delta^{40}\mathrm{Ar}_{\mathrm{atm}}$ volume fraction.

The model integrates from $t = 0$ (initial condition) up to some upper time t_1 (typically 3000 kyr), advancing with a forward Euler step $dt = 0.2\,\mathrm{kyr}$ (with an automatic retry at $dt/2$ if the integrator produces non-physical, negative concentrations).

1.3 Mixing rate profile

The depth-dependent mixing rate is parameterised by two values m_{clean} and $m_{\mathrm{dirty}} = f_{\mathrm{dirty}}\, m_{\mathrm{clean}}$ joined by a smooth transition of thickness δ centred at z_{lim}. Using w_1 and w_2 for the two tanh-shaped step functions,

$$w_1(z) = \tfrac{1}{2}\Big(1 + \tanh\big(\tfrac{\beta}{\delta}(z - z_{\mathrm{lim}})\big)\Big)\,, \tag{1}$$

$$w_2(z) = \tfrac{1}{2}\Big(1 + \tanh\big(\tfrac{\beta}{\delta}(z - z_{\mathrm{lim}} - \delta)\big)\Big)\,, \tag{2}$$

$$\Phi(z) = m_{\mathrm{dirty}}\, w_1(z)\, w_2(z) + m_{\mathrm{clean}}\, w_1(z)\,(1 - w_2(z)), \tag{3}$$

with sharpness $\beta = 50$ (essentially a step on the discretisation scale). So $\Phi = 0$ in clean ice, $\approx m_{\mathrm{clean}}$ in the transition zone of width δ, and $\approx m_{\mathrm{dirty}}$ in the dirty zone below.

The diffusivity is $D(z) = \Phi(z)\, L_{\mathrm{ref}}$ with $L_{\mathrm{ref}} = 1\,\mathrm{m}$ treated as a reference length (it just rescales m_{clean}; we keep $L_{\mathrm{ref}} = 1$ and let m_{clean} carry the magnitude).

1.4 Governing equations

The continuum dynamics of $c \in \{c_{81}, c_{40}\}$ are

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial z}\left(D(z)\,\frac{\partial c}{\partial z}\right) + S_c(z,t), \tag{4}$$

discretised cell-centre to cell-centre. The source/sink term S_c differs by tracer:

- For ^{81}Kr, $S_{81}(z,t) = -\lambda\, c_{81}(z,t)$ with $\lambda = \ln 2/t_{1/2}$ and $t_{1/2} = 229\,\mathrm{kyr}$, except in the clean-ice cells for times $t > t_{\mathrm{old}}$, where the decay term is switched off. (Physically: the clean ice was deposited at most t_{old} ago and so its ^{81}Kr has decayed for at most that long.)
- For $\delta^{40}\mathrm{Ar}_{\mathrm{atm}}$, $S_{40} = 0$ in the interior; at the bedrock cell only, a flux F_{40} is added: $S_{40}(z_{\mathrm{bed}}, t) = F_{40}/\Delta z_{\mathrm{bed}}$.

The observable quantities derived from the state are

$$\tau_{81}(z,t) = -\,t_{1/2}\,\log_2\!\big(c_{81}(z,t)\big), \qquad \text{Kr-81 closed-system age, kyr,} \tag{5}$$

$$\delta^{40}\mathrm{Ar}_{\mathrm{atm}}(z,t) = 1000\,\left(\frac{c_{40}(z,t)}{A_{40}^0} - 1\right) - 0.066\,t, \qquad \delta^{40}\mathrm{Ar},\ \text{‰}, \tag{6}$$

where the $-0.066\,t$ term in the latter expression corrects for atmospheric $\delta^{40}\mathrm{Ar}_{\mathrm{atm}}$ growth over time (Bender et al. 2010).

1.5 Parameters

The five sampled parameters and the two fixed observation-noise scales are:

Symbol	Meaning	Units	Prior	Code name
δ	transition-zone thickness	m	Uniform(0.3, 2.0)	`delta`
m_{clean}	mixing rate in transition zone	m/kyr	Normal(0.03, 0.002) trunc. $\geq$ 0	`m_clean`
f_{dirty}	dirty-to-clean mixing-rate ratio	—	Uniform(4.0, 6.5)	`f_dirty`
t_{old}	clean-ice maximum age	kyr	Normal(250, 25) trunc. ≥ 0	`t_old`
F_{40}	bedrock $\delta^{40}\text{Ar}_{\text{atm}}$ source flux	cc/m^2/kyr	Uniform(0.004, 0.007)	`F_ar40`
σ_{81}	^{81}Kr age obs. noise (*fixed*)	kyr	30	`σ_k81`
σ_{40}	δ^{40}Ar obs. noise (*fixed*)	‰	0.03	`σ_dar40`

The choice to fix σ_{81} and σ_{40} rather than place priors on them was made to improve sampling efficiency. Both values are close to the analytical errors of the measurements; relaxing them to half-Cauchy or exponential priors is a minor change to the model if needed.

1.6 Observations

The model is compared against two data sets, both at observation depths z_i corresponding to specific GISP2 ice samples:

- Three ^{81}Kr closed-system ages, τ_{81}^{obs}, at depths 3045–3050 m (typically reported as ages of 577, 649, and 714 kyr).
- Ten $\delta^{40}\text{Ar}_{\text{atm}}$ measurements from Bender et al. (2010), spanning depths 3036.5 to 3052.4 m, with values rising from ~ 0 ‰ in the clean zone to ~ 0.5 ‰ near bedrock.

For depths between cell centres the model predictions are linearly interpolated. Both data sets are assumed to be independent Gaussian observations of the corresponding model quantities at the depths in question.

2 The likelihood and the time-of-comparison problem

For given parameters $\theta = (\delta, m_{\text{clean}}, f_{\text{dirty}}, t_{\text{old}}, F_{40})$ and integration time t, the model produces predictions $\hat{\tau}_{81}(z_i;\theta,t)$ and $\widehat{\delta^{40}\text{Ar}}(z_j;\theta,t)$ at the observation depths z_i, z_j. Conditional on θ and t, the log-likelihood is the usual Gaussian:

$$\log L(t \mid \theta, \mathbf{y}) = -\frac{1}{2\sigma_{81}^2}\sum_i \left(\hat{\tau}_{81}(z_i;\theta,t) - y_i^{81}\right)^2 - \frac{1}{2\sigma_{40}^2}\sum_j \left(\widehat{\delta^{40}\text{Ar}}(z_j;\theta,t) - y_j^{40}\right)^2 + C, \quad (7)$$

where C collects θ- and t-independent normalising constants.

The wrinkle is that we have no direct information about the integration time t that should be used to compare the model with the measurements. Physically, t is the amount of time the mixing process at the base of the ice sheet has been operating before the present day. The data alone don't pin this number down: once the mixing column has run long enough to reach a steady state, all later times look identical, so any $t \geq t_{\text{eq}}(\theta)$ produces the same predictions and therefore the same likelihood.

This non-identifiability is the central reason the three sampling strategies below differ. They are all valid pieces of inference machinery but answer different questions.

3 Sampling strategies

The repository implements three strategies, selectable by the `model_kind` variable at the top of `run_basalmixing_ensemble.jl`. All three use the same affine-invariant ensemble sampler (`Emcee`; Goodman & Weare 2010) with 32 walkers, so their wall-clock costs are directly comparable per integration of the forward model.

3.1 `:profile` — profile likelihood

What it does. For every parameter proposal θ, integrate the forward model from $t = 0$ to $t_1 = 3000\,\text{kyr}$ on a 1-kyr grid. At each grid point compute the log-likelihood (7). Select

$$t^\star_{\text{pred}}(\theta) = \arg\max_t \log L(t \mid \theta, \mathbf{y}), \tag{8}$$

and use $\log L^\star(\theta) = \log L(t^\star_{\text{pred}} \mid \theta, \mathbf{y})$ as the contribution to the MCMC log-posterior.

Assumptions.

- The "correct" time t_{pred} for each θ is treated as a deterministic function of θ chosen to make the data fit best.
- No prior is placed on t_{pred}.

What you get out. A posterior on θ from a non-standard (non-Bayesian) likelihood: the *profile likelihood* $L^\star(\theta)$. The reported $t^\star_{\text{pred}}$ per chain sample is the maximizer, not a sample from a posterior.

Pros and cons. Computationally efficient because the inner integration can early-terminate once predicted ages exceed the range of interest. Single-pass per sample, no extra dimension to mix over. *However:* the posterior on θ is biased — it over-credits each θ for the best t available rather than averaging the data's support over plausible t. The reported t_{pred} "posterior" is meaningless as a probability distribution; it's a histogram of point optima.

3.2 `:sampled` — explicit t_{pred} prior, joint $(\theta, t_{\text{pred}})$ posterior

What it does. Treat t_{pred} as an additional sampled parameter, with a uniform prior $t_{\text{pred}} \sim \text{Uniform}(700, 3000)\,\text{kyr}$. For each $(\theta, t_{\text{pred}})$ proposal, run the forward model once from $t = 0$ to $t = t_{\text{pred}}$ and evaluate (7) at $t = t_{\text{pred}}$.

Assumptions.

- t_{pred} is a meaningful physical quantity that we have prior belief about (uniform on $[700, 3000]$).
- The integration finish time *is* the comparison time.

What you get out. A proper joint Bayesian posterior on $(\theta, t_{\text{pred}})$. The marginal on θ accounts for uncertainty in t_{pred} by averaging; the marginal on t_{pred} is the data's combined credibility about "how long has this been running."

Pros and cons. Bayesianly sound at the level of the joint posterior, and on average *cheaper* per evaluation than `:profile` because most proposed $t_{\rm pred}$ are less than 3000 (typical mean $t_{\rm pred} \approx 1500$ kyr, so a $\sim 2\times$ speed-up).

But the posterior on $t_{\rm pred}$ is essentially unidentifiable. Once the system has equilibrated for the proposed θ, every later t gives the same likelihood; the data don't choose between $t_{\rm pred} = 1500$ and $t_{\rm pred} = 2500$ when $t_{\rm eq}(\theta) = 1000$. The marginal $p(t_{\rm pred} \mid \mathbf{y})$ in the chain therefore:

- Has a tall peak near the smallest equilibration time that is compatible with the data (typically ~1000 kyr), where the likelihood first plateaus;
- Rises monotonically toward the upper prior bound $t_1 = 3000$ kyr. This rise is the mixture-of-uniforms effect: chain samples with smaller $t_{\rm eq}(\theta)$ contribute mass to all t above their $t_{\rm eq}$, so the higher t values accumulate contributions from more chain samples. It is a mathematically correct posterior under a Uniform$(700, 3000)$ prior, but the precise rise rate and the location of the upper edge are determined by the *arbitrary* choice of t_1, not by the data.

If one extended the prior to Uniform$(700, T_{\max})$ for some $T_{\max} > 3000$, the same rising shape would extend to $T_{\max}$ and pile up there. The boundary is artificial.

3.3 `:marginal` — treat $t_{\rm pred}$ as a nuisance parameter, integrate it out

What it does. Recognise that $t_{\rm pred}$ is not a quantity we actually care about (it has no independent physical observation) and is unidentifiable past $t_{\rm eq}(\theta)$. Treat it as a nuisance parameter and marginalise it out of the likelihood:

$$\log p(\mathbf{y} \mid \theta) \;=\; \log \int_{t_0}^{t_1} L(t \mid \theta, \mathbf{y})\, p(t)\, dt. \tag{9}$$

With a uniform prior $p(t) = 1/(t_1 - t_0)$ on $[t_0, t_1]$, the normalisation is constant in θ and can be absorbed into the chain's normalising constant. Numerically we evaluate (9) as a trapezoidal sum on the 1-kyr prediction grid:

$$\log p(\mathbf{y} \mid \theta) \;\approx\; \log \left(\sum_k \exp\big(\log L(t_k \mid \theta)\big)\, \Delta t \right), \tag{10}$$

computed via the standard `logsumexp` trick to avoid underflow.

Equilibration speed-up. For typical θ the integration spends most of its wall-clock in the post-equilibration phase, where the integrand is constant. We detect equilibration online by comparing the predicted observation vectors W slices apart (default $W = 100$ kyr) and declaring equilibrium when both ^{81}Kr and $\delta^{40}\mathrm{Ar}_{\rm atm}$ predictions have moved less than tight tolerances ($\sim 1\%$ of the corresponding observation noise scale). When equilibrium is declared at slice $k_{\rm eq}$ with time $t_{\rm eq}$ and log-likelihood $\log L_{\rm eq}$, the remainder of the integral is added in closed form:

$$\log p(\mathbf{y} \mid \theta) \;\approx\; \log \left(\sum_{k \le k_{\rm eq}} \exp\big(\log L(t_k)\big)\, \Delta t \;+\; (t_1 - t_{\rm eq})\, \exp\big(\log L_{\rm eq}\big) \right). \tag{11}$$

In practice the integrator equilibrates around $t \in [1\,000, 1\,500]$ kyr, giving roughly a 2–3$\times$ speed-up per evaluation.

Assumptions.

- t_{pred} is a nuisance parameter — we are not interested in sampling its distribution.
- The prior on t_{pred} is uniform over $[t_0, t_1]$. The choice of t_1 enters the log-likelihood only as an additive constant (the normalisation $-\log(t_1 - t_0)$), which factors out of the chain's relative weights; *the posterior on θ does not depend on t_1*, provided t_1 is large enough to be past equilibration for the bulk of the posterior support. This is the key qualitative improvement over `:sampled`.
- Equilibration is detected within tolerance $\sim 1\%$ of σ_{81} and σ_{40}; if for some θ the system equilibrates more slowly than the integration grid runs, the explicit sum (10) is used and remains correct.

What you get out. A clean marginal posterior on θ (5-dimensional, vs. 6-dimensional for `:sampled`). **There is no posterior on t_{pred} in the chain.** If a posterior on t_{pred} is needed for diagnostics, it can be reconstructed *post hoc* from the chain:

$$p(t_{\text{pred}} \mid \mathbf{y}) \;\propto\; \int p(t_{\text{pred}} \mid \mathbf{y}, \theta)\, p(\theta \mid \mathbf{y})\, d\theta \;\approx\; \frac{1}{N}\sum_{s=1}^{N} \frac{L(t_{\text{pred}} \mid \theta^{(s)})}{\int L(t \mid \theta^{(s)})\, dt}, \tag{12}$$

which averages the per-slice likelihood curves computed during sampling. The result is meaningful (it summarises what the data say about t_{pred} in light of the parameter posterior) without injecting the boundary-pile-up artefact of `:sampled`.

Pros and cons. This is the correct Bayesian treatment of an unconstrained nuisance parameter and is the recommended default. Wall-clock per evaluation is comparable to `:profile` (the integrator runs to t_{eq} rather than to t_1); the chain is one dimension smaller than `:sampled` so Emcee converges with slightly fewer iterations for the same effective sample size.

The main caveat is that the answer is qualitatively different from `:profile` or `:sampled` — the marginal posterior on θ will generally be broader because we no longer over-credit a fitted t.

4 Practical considerations

4.1 Sampler and threading

All three model kinds use Emcee with 32 walkers and 500 iterations ($32 \times 500 = 16\,000$ likelihood evaluations). Emcee is appropriate here because the posterior is moderately multimodal and the forward model is not differentiable in a way that supports gradient-based samplers (it has a `try/catch` on `DomainError` and mutates buffers, breaking `ForwardDiff`).

Threading: each thread owns its own `BasalMixingModel` object, allocated to length `Threads.maxthreadid(`
to accommodate Julia's interactive threads in newer versions.

4.2 Numerical robustness

The forward Euler integrator at $\Delta t = 0.2\,\text{kyr}$ is marginally stable; for some proposals the explicit step overshoots and produces non-positive concentrations, raising `DomainError` in `concentration_to_age`. The forward function catches these and reports failure; the `@model` retries once at $\Delta t/2$ before giving up and attaching a vanishingly small likelihood.

4.3 Outputs and reproducibility

Each ensemble run writes a snapshot to `results/emcee-chain-<model_kind>.jld2`, containing the chain, the data DataFrames, the depth grid, the priors, the sampler and the model kind. The plotting helpers (`plot_basalmixing_ensemble.jl`) load this snapshot and re-derive figures, switching the MAP best-fit re-run on the recorded `model_kind`.

References

- Bender, M. L. et al. "The Contemporary Degassing Rate of ^{40}Ar from the Solid Earth," *PNAS* 105 (24), 8232–8237 (2008); and the GISP2 measurements used here, archived with the Bender 2010 data set in this repository.

- Goodman, J. and Weare, J. "Ensemble Samplers with Affine Invariance," *Comm. App. Math. Comp. Sci.* 5, 65–80 (2010).

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